Adam Rust

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EDUCATION

University of Wisconsin-Madison | Madison, WI *May 2027*

Bachelor of Science: Computer Engineering

- GPA: 3.408 Cumulative.
- Awards: *Dean's List (2 Semesters), Deans Honor List (1 Semester).*
- Relevant Coursework: Signals Processing, Circuit Analysis, Digital Systems, Electrodynamics, Microprocessor Systems, Machine Organization.

TECHNICAL PROJECTS

FPGA-Based Hardware Control *Aug 2025 – Dec 2025*

- Implemented multiple finite state machines (FSMs) utilizing SystemVerilog to manage complex operational states.
- Configured state storage using D-Flipflops and synchronous logic to ensure stable state transitions.
- Developed a 7-segment decoder using combinational logic to translate 4-bit truth tables into a visual numeric display.
- Verified design functionality through comprehensive RTL simulation in QuestaSim prior to hardware deployment.

Microcontroller-Controlled LED PCB Game. *Jan 2026 – May 2026*

- Routed a circular cascaded array of 12 addressable RGB LEDs, utilizing decoupling capacitors.
- Engineered a dual-rail power system to safely drop a 5V power source to a stable 3.3V logic environment.
- Implemented a hardware-level voltage supervisor and watchdog timer to prevent software and hardware lockups.

Standalone Custom 8-Bit Digital Calculator *Aug 2026 – Present*

- Developed schematics and a 2-layer PCB layout using a direct-pin I/O keypad array, an ATtiny-1617 microcontroller, and a dedicated SPI 7-segment display.
- Integrated test points across the board layout to allow for real-time signal debugging and voltage validation.

Embedded RTOS Application *Jan 2026 – May 2026*

- Developed a multi-task embedded application using FreeRTOS on ARM Cortex-M, implementing task synchronization via queues, semaphores, and event groups.
- Implemented serial communication protocols (UART, I2C, SPI) to interface with external peripherals.
- Designed interrupt-driven I/O using NVIC priority configuration and ISRs utilizing volatile global variables.

Low-Level Software Architecture *Aug 2024 – May 2025*

- Created multiple methods in x86-64 assembly managing register allocation, stack frames, and calling conventions.
- Utilized bitwise operations and hexadecimal arithmetic to manipulate data within limited register storage.

TECHNICAL SKILLS

Languages: SystemVerilog (HDL), C/C++, Java, Python, Bash, Assembly, HTML/CSS.
Tools: QuestaSim, Quartus, MATLAB, LTspice, Make, Git/GitHub, Altium Designer, FreeRTOS.
Hardware & Lab: PCB Design, Mixed-Signal Routing, Oscilloscope, Digital Multimeter, Soldering.

INVOLVEMENT / LEADERSHIP

University of Wisconsin Marching Band | UW-Madison | Madison, WI *Aug 2023 – Present*

Saxophone Rank Leader

- Selected as Rank Leader to spearhead the structural organization and placement of members within the saxophone section for high-profile field performances.
- Directed instructional blocks for the saxophone section, mentoring individuals on high-precision marching techniques and strict drill coordination to ensure peak performance standards.
- Facilitated a teamwork-oriented environment and adapted leadership styles across diverse peer groups.